Functions in the form (ax + b) to a po[redacted]

1

a $\dfrac{(x-5)^5}{5}+C$

b $-\dfrac{(x+6)^{-2}}{2}+C$

c $2(x+2)^3+C$

d $-2(x-5)^{-2}+C$

e $\dfrac{(2x+3)^4}{8}+C$

f $-\dfrac{(4x-3)^{-3}}{12}+C$

g $-\dfrac{(2x-3)^{-3}}{6}+C$

h $-\dfrac{(4x+3)^{-1}}{4}+C$

i $\dfrac{(2x-5)^{\frac{3}{2}}}{3}+C$

j $\dfrac{(4x+5)^{\frac{1}{2}}}{2}+C$

k $\dfrac{(4x+5)^{\frac{9}{4}}}{9}+C$

l $-\dfrac{(1-x)^7}{7}+C$

m $-(5-x)^4+C$

n $-\dfrac{(4-3x)^7}{21}+C$

o $\dfrac{(2x+5)^4}{2}+C$

p $-\dfrac{(2-3x)^4}{3}+C$

q $\dfrac{(4-3x)^{-4}}{12}+C$

r $\dfrac{5(5-4x)^{-2}}{8}+C$

s $-\dfrac{(7-4x)^{\frac{3}{2}}}{6}+C$

t $(2x+3)^{\frac{3}{2}}+C$

u $-\dfrac{5(1-4x)^{\frac{1}{2}}}{2}+C$

v $2(6x+5)^{\frac{4}{3}}+C$

w $5x+\dfrac{(4x+1)^4}{16}+C$

x $10(x-2)^{\frac{1}{2}}+C$

y $\dfrac{2(x-2)^{\frac{3}{2}}}{3}+\dfrac{2(x+3)^{\frac{3}{2}}}{3}+C$

2 $y=\dfrac{3}{2}(2x-1)^4-\dfrac{1}{2}$

3 $y=\dfrac{(5x-6)^3}{3}+\dfrac{4}{3}$

4

a $y=\dfrac{\sqrt{(6x+66)^3}}{9}-7$

b $y=\dfrac{(4x-2)^4}{16}-2$

Functions in the form f'(x).f(x)

5

a $10x\left(x^2-3\right)^4$

b $3\left(x^2-3\right)^5+C$

6

a $50\left(5x^2+10x-3\right)^4(x+1)$

b $3\left(5x^2+10x-3\right)^5+C$

7

a $5\left(x^2-3x+6\right)^4(2x-3)$

b $-\frac{1}{5}\left(x^2-3x+6\right)^5+C$

8

a $\frac{2}{\sqrt{4x+11}}$

b $6\sqrt{4x+11}+C$

9

a $-\frac{6x}{\left(x^2+7\right)^4}$

b $\frac{4}{\left(x^2+7\right)^3}+C$

10

a $3x^2e^{x^3}$

b $e^{x^3}+C$

11 $3e^{3x^4-5}+C$

12 $\frac{1}{5}\sin\left(x^5\right)+C$

13 $-5\cos\left(x^6+3x^5\right)+C$

14 $8e^{\sqrt{x}}+C$

15

a $2\left(3x^4+2x\right)^7+C$

b $-\frac{1}{6}\cos^2 3x+C$

c $2e^{\sin x}+C$

d $\frac{1}{6}e^{3x^2+12x}+C$

Integration as reverse of differentiation

16

a $-\dfrac{\sin x+\cos x}{e^x}$

b $-\dfrac{\cos x}{e^x}+C$

17

a $e^x(\sin x+\cos x)$

b $3e^x \sin x+C$

18

a $\dfrac{1}{\cos^2 x}$

b $5\tan x+C$

Rearrangement method

19

a $4\sin^3(t)=3\sin t-\sin 3t$

b $-\dfrac{9}{4}\cos t+\dfrac{1}{4}\cos 3t+C$

20

a $(1+x)e^x$

b No solution provided.

c $(x-1)e^x+C$

21

a $\dfrac{1-x}{e^x}$

b No solution provided.

c $-\dfrac{x+1}{e^x}+C$

22

a $\cos x-x\sin x$

b $\sin x+x\cos x+C$